puremoe: A PMID-Centered R Interface to PubMed and NIH/NLM Literature Data

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Abstract

Biomedical literature projects commonly begin in PubMed but require data from several additional National Institutes of Health and National Library of Medicine services. These services differ in their interfaces, response formats, batching rules, and data structures. The puremoe package provides a PMID-centered R interface for searching PubMed and retrieving article metadata, iCite metrics and citation links, PubTator3 annotations and relations, and open-access PubMed Central full text. It keeps remote retrieval explicit and provides source-specific local workflows for MeSH keyness, citation snowballing and network construction, and sentence-level context for biomedical entities and relations. We describe the package's design, compare its scope with adjacent R clients, and demonstrate the workflow with a dated corpus on buprenorphine and opioid use disorder. puremoe is intended for moderately sized literature projects that require a structured NIH/NLM-centered corpus without maintaining separate API clients or a local literature index.

Keywords: PubMed; MeSH; PubTator3; citation networks; literature retrieval; R

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1. Introduction

PubMed is the principal public gateway to the biomedical journal literature, but searching it is only one part of many literature-analysis workflows (Lu, 2011). PubMed records provide bibliographic metadata and Medical Subject Headings (MeSH), the controlled vocabulary used to index MEDLINE (Lipscomb, 2000). Complementary NIH and NLM services add other forms of evidence: iCite supplies article-level citation measures and citation links (Hutchins et al., 2016), PubTator3 identifies biomedical entities and relations in text (Wei et al., 2024), and PubMed Central (PMC) distributes open-access full text. PMIDs connect these resources conceptually, yet the resources do not share a single programming interface.

The resulting integration work is substantial even when the scientific question is modest. Each service has distinct request conventions, batch sizes, rate limits, response formats, and data models. An R workflow may therefore need to divide the same PMID vector several ways, parse XML,

JSON, and BioC data, and preserve nested annotations without duplicating articles. Combining citation data, controlled vocabulary, and biomedical annotations requires both reliable retrieval and representations that remain usable after retrieval.

`puremoe` (Timm, 2026) treats this task as a PMID-centered integration problem. Its contribution is not a new literature database or search language. It turns a PubMed-derived or user-supplied PMID set into reusable R objects that can be enriched across selected NIH/NLM services and passed into local workflows without hidden retrieval. A compact interface covers PubMed, iCite, PubTator3, and PMC (Roberts, 2001); the corresponding local workflows include MeSH keyness, citation-neighborhood expansion and network construction, and sentence-level context for biomedical entities and relations. Keeping retrieval and local computation separate makes the intermediate objects available for inspection, saving, joining, or analysis with ordinary R tools.

The package also ships a PubMed-wide snapshot of MeSH descriptor frequencies, which supplies the background distribution needed to distinguish descriptors that characterize a selected corpus from headings that are merely common throughout PubMed.

This article positions `puremoe` among adjacent R clients, describes its interface and data model, and demonstrates the workflow with a corpus on buprenorphine and opioid use disorder. The final sections discuss implementation, performance, reproducibility, and the limits of the package's deliberately NIH/NLM-centered scope.

2. Existing approaches in R

R already has capable clients for several parts of this workflow. `rentrez` provides a general interface to NCBI E-utilities and supports the many databases in the Entrez system (Winter, 2017). `easyPubMed` (Fantini, 2025) and `pubmedR` (Aria, 2026) focus more directly on PubMed retrieval and parsing; `pubmedR` also provides citation-oriented helpers and data that can enter a `bibliometrix` workflow. These packages are natural choices when PubMed records, rather than coordinated access to several NIH/NLM services, are the primary object of interest.

Two broader scholarly-data clients provide important points of comparison. `europepmc` (Jahn & Salmon, 2023) exposes the Europe PMC search index, references and citing works, annotations, and available full text; the underlying service integrates records from multiple life-science sources (Ferguson et al., 2021). `openalexR` targets the cross-disciplinary OpenAlex scholarly graph and includes operations for retrieving citation neighborhoods (Aria et al., 2024). Their broader indexes can be preferable to a PubMed-centered workflow when source breadth or cross-domain citation analysis is the main requirement.

The overlap among these packages does not make their operations equivalent. Europe PMC annotations are not identical to direct PubTator3 output, an OpenAlex open-access location is not parsed PMC XML, and generic access to an Entrez endpoint does not by itself provide a MeSH keyness calculation or an evidence-bearing entity relation network. Table 1 therefore records operations exposed as native package workflows. "Generic" denotes a task that can be constructed

through a lower-level interface, rather than a dedicated operation with the same returned structure.

Table 1. Native, documented capabilities in puremoe and adjacent R packages. Documentation was audited for puremoe 1.1.1, rentrez 1.2.4, easyPubMed 3.1.6, pubmedR 1.0.2, europepmc 0.4.3, and openalexR 3.1.0.

capability	puremoe	rentrez	easyPubMed	pubmedR	europepmc	openalexR
Direct PubMed/NCBI search	Native	Native	Native	Native	Partial	—
PubMed abstracts and metadata	Native	Generic / partial	Native	Native	Native	Partial
Retrieve an existing PMID set	Native	Native	Native	Native	Native	Partial
NIH iCite metrics and links	Native	—	—	—	—	—
PubTator3 entities and relations	Native	—	—	—	Partial	—
PMC open-access full text	Native	Generic / partial	—	—	Native	Partial
MeSH terms in article records	Native	Generic / partial	Native	Native	Native	—
MeSH thesaurus and tree data	Native	Generic / partial	—	—	—	—
References and citing works	Native	Generic / partial	Partial	Native	Native	Native
Citation snowballing helper	Native	—	—	—	—	Native
Ready-to-use citation network	Native	—	—	—	—	Partial
MeSH corpus keyness	Native	—	—	—	—	—
Sentence-level entity context and evidence network	Native	—	—	—	—	—

Several qualifications are important when reading Table 1. `europepmc` searches the Europe PMC index, which includes MEDLINE records, rather than sending a query to NCBI PubMed. Its annotations are related text-mining data but are not a direct PubTator3 client. `rentrez` can reach PMC, MeSH, and NCBI link sets through the generic Entrez interface, but users must choose the databases and link types and then perform the corresponding parsing and analysis. `easyPubMed` can extract references present in PubMed XML; this is not a general cited-by service. Finally, OpenAlex works can carry PMIDs and open-access locations, but `openalexR` is neither a PubMed client nor a PMC XML parser; its snowballing output requires a further step to become a graph.

The comparison is therefore a statement of scope, not a claim that `puremoe` replaces these clients. Its narrower contribution is a consistent path through selected NIH/NLM literature services followed by local workflows designed for the resulting objects. Broader NCBI or scholarly-graph questions remain better served by `rentrez` or `openalexR`, while Europe PMC-centered work is naturally handled by `europepmc`.

3. Package design

Three choices organize the package: PMIDs provide the common key, remote retrieval is explicit, and one compact dispatcher exposes endpoint-specific operations. Retrieved objects then enter source-specific local workflows rather than a general analysis framework.

3.1 PMIDs as the common key

PMIDs are the organizing key rather than an incidental field. Users may begin with `search_pubmed()` or supply identifiers collected elsewhere. The same character vector is then reusable across PubMed, iCite, and PubTator3. PMC is the one qualified case: PMIDs are first resolved to open-access XML locations. This organization minimizes identifier reconciliation and makes joins across returned tables explicit. Figure 1 summarizes the data flow: remote services enrich a common corpus, normalized objects form the boundary of the retrieval layer, and local workflows consume those objects.

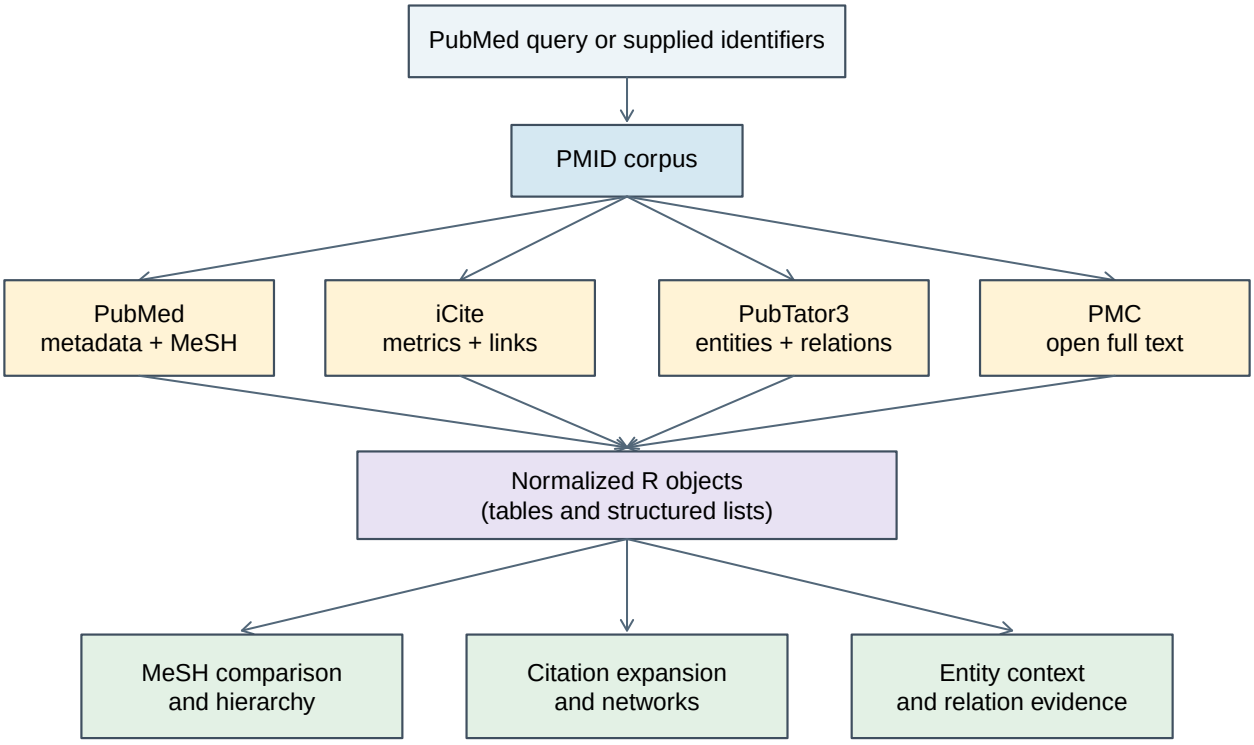


Figure 1. PMID-centered package architecture. Solid arrows show retrieval and transformation; the lower row contains local workflows that make no additional service requests.

3.2 Retrieval and local workflows

The public retrieval layer consists primarily of `search_pubmed()` and `get_records()`. The former resolves a PubMed query to PMIDs; the latter dispatches a PMID vector or endpoint-specific identifiers to a selected source. Endpoint-specific batching, requests, and parsing remain behind this shared surface. The returned tables and lists then enter local workflows designed for their source structures, without making additional API requests (Timm, 2026).

Each local workflow follows the structure of its source. `mesh_keyness()` uses retrieved PubMed records, `citation_network()` uses an iCite result already in memory, and the PubTator3 context functions use a previously returned entity/relation object. These functions accept retrieved data and make no further service requests. They return tables that users can save, filter, join, or pass to other R tools. Users can therefore revise a threshold, network, or display without sending new requests to an upstream service.

3.3 Endpoint schemas

The supported endpoints differ in both their accepted inputs and returned objects. Table 2 is generated from `endpoint_info()` rather than maintained as a separate handwritten schema, so it describes the installed package

version.

Table 2. Remote endpoints exposed by the shared retrieval function.

endpoint	source	input	returns
pubmed_abstracts	PubMed E-utilities	PMIDs	data.table; one row per PMID
pubmed_affiliations	PubMed E-utilities	PMIDs	data.table; one row per author-affiliation record
icites	NIH iCite	PMIDs	data.table; one row per PMID returned by iCite
pubtator	PubTator3 BioC JSON export	PMIDs	list with entities and relations data.tables
pmc_fulltext	PMC Cloud Service XML files	HTTPS XML URLs, usually from pmid_to_ftp()$url	data.table; one row per parsed top-level body section

As Table 2 shows, the `endpoint` argument keeps the public retrieval surface small while allowing the implementation to use service-specific request and parsing strategies. `endpoint_info()` makes those differences inspectable: for each endpoint it reports the upstream source, accepted input, returned object, relevant parameters, rate-limit guidance, and endpoint-specific notes. This metadata is part of the interface rather than a substitute for it; returned objects use stable package-level column names even though the remote representations differ.

4. Worked example: Buprenorphine and opioid use disorder

The example corpus contains PubMed records explicitly centered on buprenorphine and opioid use disorder (OUD). Requiring both concepts in article titles supplies a transparent precision rule and keeps the live workflow manageable. Buprenorphine is well established in maintenance treatment, and comparisons with methadone and other treatment strategies provide a literature with chemical, disease, intervention, and outcome concepts (Mattick et al., 2014). The example follows one corpus through a connected sequence: PubMed defines it, MeSH characterizes it, iCite tests what its title wording leaves outside, PubTator3 makes predicted relations inspectable in text, and PMC supplies selected full text for deeper reading. The query does not define a systematic-review search: in particular, its contemporary title wording underrepresents older literature described with terms such as *opioid dependence*.

4.1 Corpus definition and article metadata

The workflow first defines a transparent seed set in a reusable form. It begins with the literal PubMed query; `search_pubmed()` returns the PMID vector that serves as the input to every subsequent enrichment.

```
query <- 'buprenorphine[Title] AND "opioid use disorder"[Title]'
pmids <- puremoe::search_pubmed(query)
```

The query returned 784 identifiers on 2026-09-11. Recording both the literal query and date is necessary because PubMed is a changing index. The example intentionally favors a transparent selection rule over

maximal recall; later citation expansion provides an empirical view of what this narrow wording leaves outside the seed set.

The seed PMIDs are passed to the shared retrieval function with the PubMed metadata endpoint. The parallelism and delay are explicit retrieval parameters, not global package state. Table 3 shows five returned records and the principal article-level fields used later in the workflow.

```
abstracts <- puremoe::get_records(
  pmids,
  endpoint = "pubmed_abstracts",
  cores = 3L,
  sleep = 1.05
)
```

Table 3. Example PubMed records returned by the shared retrieval function.

pmid	year	journal	articletitle
42709580	2026	Journal of addiction medicine	Treatment Discontinuation and Barriers and Facilitators to Retention Among Individuals Initiating Buprenorphine for Opioid Use Disorder: A Longitudinal Cohort Study.
42644692	2026	The American journal of drug and alcohol abuse	Extended-release naltrexone versus buprenorphine-naloxone for opioid use disorder: a systematic review and meta-analysis of randomized controlled trials.
42642688	2026	AIDS and behavior	Trends in Buprenorphine Prescribing Among Adults with Opioid Use Disorder and Coexisting HIV or Viral Hepatitis, 2015-2025.
42613192	2026	Substance use & addiction journal	Toward a Pharmacy-Led Value-Based Care Model for Buprenorphine Initiation and Management (PharmValue): Development of a Scalable Pharmacist-Led Collaborative Practice Agreement for Medications for Opioid Use Disorder.
42607586	2026	Drug and alcohol dependence	Predictors of incident anxiety and stress-related diagnoses following buprenorphine initiation among commercially insured patients with opioid use disorder.

In Table 3, each returned article occupies one row, while nested annotations such as MeSH descriptors are retained in list-columns rather than flattened into duplicated article rows. This representation preserves an article-level unit for joins and is also the input subsequently consumed by `mesh_keyness()`.

The same seeds can acquire complementary evidence without identifier reconciliation. Changing only `endpoint` enriches them with citation and biomedical text-mining data. Keeping both calls together emphasizes the common retrieval pattern and the reuse of one identifier vector.

```
icites <- puremoe::get_records(
  pmids,
  endpoint = "icites",
  cores = 3L,
  sleep = 0.25
)
pubtator <- puremoe::get_records(
  pmids,
  endpoint = "pubtator",
  cores = 3L,
  sleep = 0.34
)
```

The iCite result retains one row per seed returned by the upstream service, with metrics and citation links in article-level fields and list-columns; it becomes the input to both expansion and within-corpus network construction. In the reported snapshot, PubMed returned records for 779 and iCite for 778 of the 784 requested PMIDs; corpus percentages below use the returned PubMed records. The absent identifiers are recorded in the retrieval audit supplied with the article; `get_records()` binds the records supplied by the endpoint rather than manufacturing placeholder rows. PubTator3 instead returns BioC JSON in which passages contain nested entity annotations and a separate document-level list contains typed relation assertions. `get_records()` parses that structure into two PMID-linked tables: one row per entity mention and one row per predicted relation. Entity rows preserve labels, identifiers, offsets, and source passages, while relation rows identify their endpoints through mention indices. Sentence context is added locally later in the workflow.

4.2 MeSH descriptors and corpus keyness

MeSH characterizes the seeds in controlled-vocabulary terms that extend beyond the words used in their titles. The workflow first inspects the vocabulary attached to individual records and then compares its corpus prevalence with a PubMed-wide reference. `puremoe` supplies the vocabulary, hierarchy, and frequency resources needed for both steps as local R objects.

`puremoe` exposes the MeSH vocabulary and its tree placements as two tabular resources. The first supports term-to-descriptor lookup; the second preserves the potentially multiple hierarchical locations of each descriptor. Tables 4 and 5 show these two representations for the concepts defining the example.

```
mesh_thesaurus <- puremoe::data_mesh_thesaurus()
mesh_trees <- puremoe::data_mesh_trees()
```

Table 4. MeSH descriptors and example entry terms.

DescriptorName	DescriptorUI	entry_terms
Buprenorphine	D002047	Buprenorphine; Subutex; Temgésic; Buprex; Temgesic; Buprenorphine Hydrochloride
Opioid-Related Disorders	D009293	Opioid-Related Disorders; Opioid-Related Disorder; Opioid Related Disorders; Addiction, Opioid; Dependence, Opioid; Opiate Addiction

Table 5. Hierarchical locations of the example descriptors.

DescriptorUI	DescriptorName	tree_location	category	branch
D002047	Buprenorphine	D03.132.577.249.150	Chemicals and Drugs	Alkaloids
D002047	Buprenorphine	D03.605.497.150	Chemicals and Drugs	Heterocyclic Compounds, Bridged-Ring
D002047	Buprenorphine	D03.633.400.686.150	Chemicals and Drugs	Heterocyclic Compounds, Fused-Ring
D002047	Buprenorphine	D04.615.723.795.150	Chemicals and Drugs	Polycyclic Aromatic Hydrocarbons
D009293	Opioid-Related Disorders	C25.775.643.500	Diseases	Substance-Related Disorders
D009293	Opioid-Related Disorders	F03.900.647.500	Psychiatry and Psychology	Substance-Related Disorders

Table 4 connects preferred descriptors to entry terms and supplementary concepts. In this example, *Dependence, Opioid* appears as an entry term for *Opioid-Related Disorders*, making a terminological bridge that the literal title query does not provide. Table 5 represents hierarchical placement separately: buprenorphine occupies several chemical branches, while opioid-related disorders appears under both disease and psychiatry/psychology branches. A single descriptor-to-parent column would lose that multiplicity. Keeping both resources as ordinary tables permits direct lookup, joining, and aggregation with other R tools.

`mesh_keyness()` compares the proportion of corpus articles indexed with each descriptor against its prevalence across PubMed. By default, it uses the package's bundled `data_mesh_frequencies` snapshot, making the comparison a local operation. In the package version used for this article, that baseline was constructed in April 2026 from 39,703,112 distinct PMIDs in a local PubMed mirror; descriptor metadata came from the contemporaneous NLM MeSH thesaurus. Users may supply another frequency table when appropriate.

The baseline distinguishes descriptors characteristic of the corpus from broad headings that are frequent throughout PubMed, while `min_count` excludes very rare corpus descriptors from the ranking.

```
keyness <- puremoe::mesh_keyness(
  abstracts,
```

```
  measure = "log_odds",
  min_count = 5L
)
```

The most frequent MeSH descriptors in this archived corpus were *Humans* (626), *Opioid-Related Disorders* (623), *Buprenorphine* (589), *Opiate Substitution Treatment* (434), *Female* (322), and *Adult* (302). Frequency alone therefore mixes descriptors central to the topic with broad indexing categories. Table 6 instead reports the six descriptors ranked highest by standardized log odds, together with the keyness values and the prevalences being contrasted.

Table 6. MeSH descriptors ranked highest by standardized log odds against the PubMed-wide baseline.

Descriptor	Corpus articles	Corpus (%)	PubMed (%)	Log odds	Standardized z
Opiate Substitution Treatment	434	55.7	0.01	9.08	123.80
Buprenorphine	589	75.6	0.02	9.61	114.33
Narcotic Antagonists	292	37.5	0.04	7.28	97.83
Opioid-Related Disorders	623	80.0	0.06	8.74	97.55
Buprenorphine, Naloxone Drug Combination	107	13.7	0.00	9.20	82.62
Analgesics, Opioid	294	37.7	0.17	5.87	79.36

Table 6 elevates treatment-specific descriptors that are unusually concentrated in the corpus rather than simply common in biomedical indexing. The contrast shows how controlled-vocabulary enrichment can characterize a corpus beyond words in titles. The scores are descriptive rankings, not independent hypothesis tests or evidence about comparative treatment effectiveness.

4.3 Citation-based corpus exploration

Citation data expose connected literature omitted by the deliberately narrow title wording and show how its addition changes the corpus. The iCite result already in memory supports two local views: an auditable expansion beyond the seeds and a graph of citations within them.

Following references and citing articles is a well-established complement to database searching, especially when terminology changes or evidence crosses indexing boundaries (Greenhalgh & Peacock, 2005). Here, candidates must have at least two links to the seed neighborhood, and the total returned set is capped at 2,000 records.

```
snowball <- puremoe::citation_snowball(
  icites,
  direction = "both",
  min_links = 2L,
  max_nodes = 2000L
)
```

The configured threshold produced 778 seeds and 1222 candidates, filling the 2,000-record cap. The cap is a computational and screening decision, not a claim that the neighborhood is exhausted. Separate cited and citing counts expose why each candidate was admitted and allow alternative thresholds without another iCite request.

Candidate PMIDs can be sent back through the same PubMed metadata endpoint. Table 7 joins those records to their admission counts and lists the strongest links to the seed neighborhood. The ordinary joins, sorting, and display code are hidden because they are not additional `puremoe` operations.

Table 7. Highest-ranked articles discovered through citation links.

pmid	year	articletitle	cited_links	citing_links	link_count
28446428	2017	Mortality risk during and after opioid substitution treatment: systematic review and meta-analysis of cohort studies.	140	1	141
24500948	2014	Buprenorphine maintenance versus placebo or methadone maintenance for opioid dependence.	128	0	128
29913516	2018	Medication for Opioid Use Disorder After Nonfatal Opioid Overdose and Association With Mortality: A Cohort Study.	104	1	105
32022884	2020	Comparative Effectiveness of Different Treatment Pathways for Opioid Use Disorder.	74	4	78
25919527	2015	Emergency department-initiated buprenorphine/naloxone treatment for opioid dependence: a randomized clinical trial.	73	0	73
23961726	2014	Treatment retention among patients randomized to buprenorphine/naloxone compared to methadone in a multi-site trial.	65	0	65
12924748	2003	The Clinical Opiate Withdrawal Scale (COWS).	59	0	59
29150198	2018	Comparative effectiveness of extended-release naltrexone versus buprenorphine-naloxone for opioid relapse prevention (X:BOT): a multicentre, open-label, randomised controlled trial.	57	2	59
32511106	2020	The ASAM National Practice Guideline for the Treatment of Opioid Use Disorder: 2020 Focused Update.	54	1	55
30352671	2018	Risk factors for discontinuation of buprenorphine treatment for opioid use disorders in a multi-state sample of Medicaid enrollees.	49	5	54

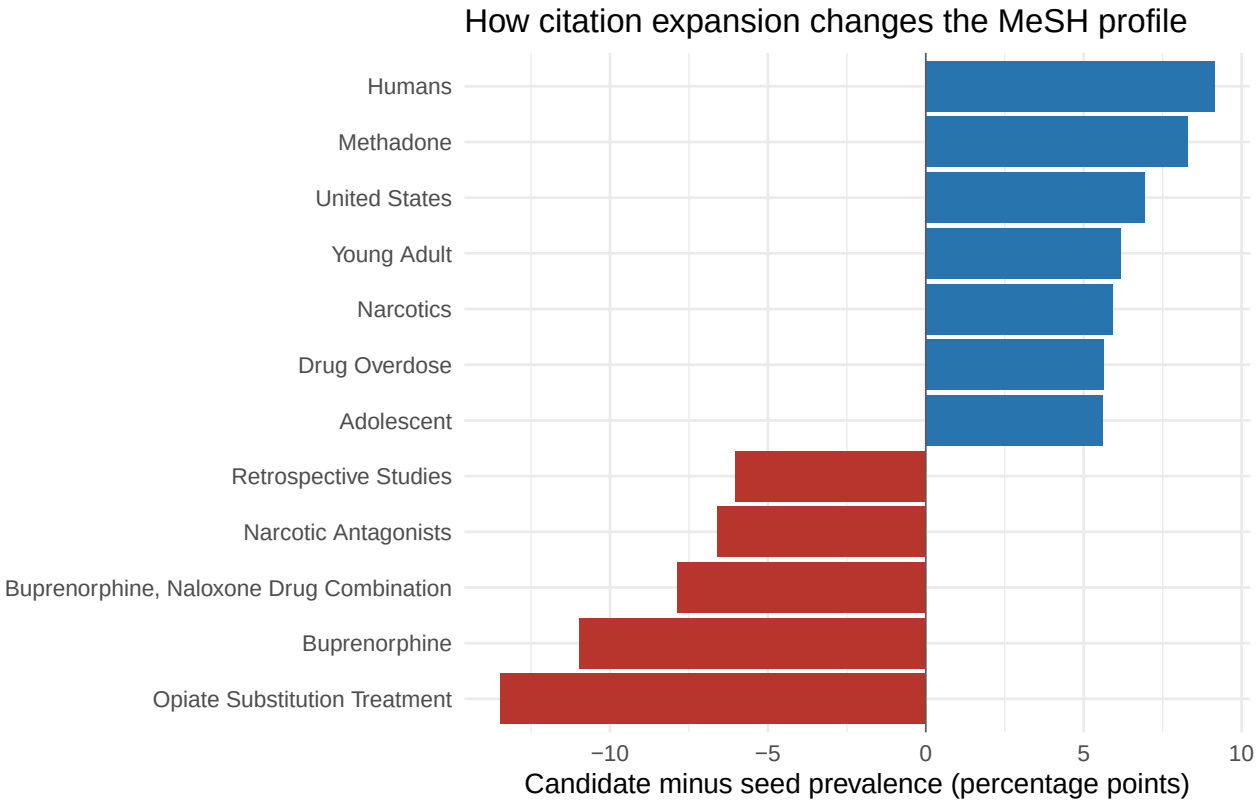


Figure 2. Difference in MeSH descriptor prevalence between citation-discovered candidates and the title-defined seed corpus. Positive values indicate higher prevalence among candidates; negative values indicate higher prevalence among seeds.

Table 7 exposes each candidate's admission counts. Several highly connected records are trials, reviews, and guidelines that predate most seeds; titles using *opioid dependence* appear prominently. Citation expansion thus recovers historically connected work that a literal title condition misses. Figure 2 complements the ranked candidates by comparing MeSH prevalence in the seeds and the citation-discovered set: positive values identify descriptors more common among candidates, while negative values identify descriptors concentrated in the seeds. Neither link rank nor vocabulary shift establishes relevance on its own, but both provide inspectable signals for subsequent screening.

The same iCite object can instead be restricted to links within the seed corpus. `citation_network()` returns conventional node and edge tables; one additional line converts them to an `igraph` object for standard graph summaries.

```
citation_net <- puremoe::citation_network(icites)
citation_graph <- graph_from_data_frame(
  citation_net$edges,
  directed = TRUE,
  vertices = citation_net$nodes
)
```

The within-corpus graph contains 778 nodes and 1907 directed edges. The presence of 107 isolates and 111

weak components is itself informative: a topically coherent query need not produce a single connected citation community. It also cautions against interpreting this graph as complete scholarly connectivity, because it contains only iCite links among the retained PubMed records. Returning node and edge tables is more useful than printing an unlabeled network; users can pass them to `igraph` (Csárdi & Nepusz, 2006) and select layouts, filters, or network statistics appropriate to the research question.

4.4 PubTator3 annotations in sentence context

Citation links test the boundary of the seed corpus; the PubTator3 workflow returns attention to concepts and predicted relations inside the retrieved texts. PubTator3 performs biomedical entity and relation extraction and returns its predictions in BioC structures (Wei et al., 2024). `get_records()` flattens those structures into linked entity mention and relation tables. `pubtator_context()` then performs a lightweight local NLP operation that attaches sentence context, after which `pubtator_cooccurrence()` and `pubtator_network()` construct corpus-level summaries. The package does not train or run a second biomedical extraction model; its contribution here is parsing, alignment, aggregation, and retention of the evidence connecting each summary back to the source PMID and sentence.

`pubtator_context()` takes the entity and relation tables returned by `get_records(endpoint = "pubtator")`. The BioC response contains passage text and document-relative annotation offsets, so the material needed to reconstruct context is present, but sentences are not supplied as an analytical table and are not attached to relation rows. The function groups passages by PMID and title/abstract location, segments each passage into sentences, and records the resulting sentence spans. It aligns each entity's document offset with a sentence span and adds a sentence identifier plus sentence-relative entity coordinates. It then resolves each relation's two mention indices against the entity table, adding endpoint text, identifiers, passage locations, and sentence identifiers. From those fields it derives `same_sentence` and `sentence_distance`. The returned `entities`, `relations`, and `sentences` tables therefore make textual proximity directly queryable without another service request or a user-written BioC and offset-alignment pipeline.

Once that context exists, `pubtator_cooccurrence()` constructs a local summary from the entity table; it does not use or expand PubTator3's predicted relation rows. It counts mention pairs within a chosen sentence window. Aggregating `by = "type"` describes broad combinations such as chemical–disease mentions, whereas `by = "entity"` retains the identifiers and mention labels supplied to the function. The latter makes normalization an explicit composition step: the article joins PubTator3 identifiers to the MeSH descriptor and supplementary concept records loaded for Table 4, while preserving the forms attested in the source text. Identifiers from other vocabularies remain unchanged.

```
context <- puremoe::pubtator_context(pubtator)
cooccurrence_types <- puremoe::pubtator_cooccurrence(
  context,
  window = 0L,
  by = "type"
)
```

```
cooccurrence_concepts <- puremoe::pubtator_cooccurrence(
  concept_entities,
  window = 0L,
  by = "entity"
)
```

Table 8. Most frequent same-sentence entity pairs after MeSH label resolution. Bracketed values are MeSH UIs; attested columns retain source forms.

type_x	concept_x	attested_x	type_y	concept_y	attested_y	n	n_pmids
Chemical	Buprenorphine [D002047]	buprenorphine; Buprenorphine; BUP	Disease	Opioid-Related Disorders [D009293]	OUD; opioid use disorder; Opioid Use Disorder	1869	723
Chemical	Buprenorphine [D002047]	buprenorphine; Buprenorphine; BUP	Species	patients	patients	716	321
Disease	Opioid-Related Disorders [D009293]	OUD; opioid use disorder; Opioid Use Disorder	Species	patients	patients	523	266
Chemical	Buprenorphine [D002047]	buprenorphine; Buprenorphine; BUP	Chemical	Naloxone [D009270]	naloxone; Naloxone; NLX	476	167
Chemical	Buprenorphine [D002047]	buprenorphine; Buprenorphine; BUP	Chemical	Methadone [D008691]	methadone; Methadone; MET	431	130
Chemical	Naloxone [D009270]	naloxone; Naloxone; NLX	Disease	Opioid-Related Disorders [D009293]	OUD; opioid use disorder; Opioid Use Disorder	259	129
Chemical	Methadone [D008691]	methadone; Methadone; MET	Disease	Opioid-Related Disorders [D009293]	OUD; opioid use disorder; Opioid Use Disorder	228	104
Chemical	Buprenorphine [D002047]	buprenorphine; Buprenorphine; BUP	Species	patient	patient	150	106
Chemical	Methadone [D008691]	methadone; Methadone; MET	Chemical	Naloxone [D009270]	naloxone; Naloxone; NLX	112	34
Disease	Opioid-Related Disorders [D009293]	OUD; opioid use disorder; Opioid Use Disorder	Disease	Substance-Related Disorders [D019966]	addiction; use; SUD	108	73
Chemical	Buprenorphine [D002047]	buprenorphine; Buprenorphine; BUP	Chemical	Naltrexone [D009271]	naltrexone; Naltrexone; NTX	108	35
Chemical	Buprenorphine [D002047]	buprenorphine; Buprenorphine; BUP	Disease	Pain [D010146]	pain; Pain; painful serious illness	97	48

Chemical–disease pairs dominate same-sentence co-occurrence in this corpus, appearing 5828 times across 734 PMIDs. For Table 8, forms such as *OUD* and *opioid use disorder* therefore resolve to the preferred MeSH label *Opioid-Related Disorders* when PubTator3 supplies the corresponding UI. Replacing the working copy's mention label before pairing also de-duplicates multiple forms of one mapped concept within a sentence; the original forms remain visible in `attested_x` and `attested_y`. Unmapped entity types retain their PubTator3 labels and identifiers rather than being forced into MeSH. Because both contextualization and the vocabulary join are local, users can change the sentence window or inspect unmapped rows without another PubTator3 call. Co-occurrence should be interpreted as proximity in text, not as a biomedical association.

Relations are aggregated only after sentence context has been attached. `pubtator_network()` resolves each endpoint to a node keyed by the PubTator3 concept identifier when available, with a type-scoped surface form as the fallback. Repeated assertions with the same directed endpoints and relation type collapse into one edge carrying total assertion, distinct-PMID, and same-sentence support counts. A separate evidence table preserves one row per original PubTator3 relation, including its PMID, relation identifier, endpoint distance, and source sentence when both mentions map to that sentence. The function therefore does not predict new relations; it converts PubTator3's document-level assertions into graph-ready corpus summaries without discarding the rows needed to audit them. Table 9 displays evidence rows for relations whose endpoints were mapped to the same sentence.

```
relation_net <- puremoe::pubtator_network(context)
```

Table 9. Sentence-level evidence retained for PubTator3 relations after MeSH label resolution. Mention columns preserve the source text.

from_concept	from_mention	to_concept	to_mention	relation_type	pmid	sentence
Buprenorphine [D002047]	BUP	Opioid-Related Disorders [D009293]	OUD	negative correlation	41063335	CONCLUSIONS: Overall, the 8-hour intravenous 1200 mcg BUP micro-infusion protocol was well-tolerated...
Buprenorphine [D002047]	buprenorphine	Opioid-Related Disorders [D009293]	OUD	negative correlation	41100863	Our data provide early evidence for safety, adherence, and reduced substance use among perinatal pat...
Buprenorphine [D002047]	Buprenorphine	Opioid-Related Disorders [D009293]	Opioid Use Disorder	negative correlation	41137494	Trends in Initiation of Buprenorphine for Opioid Use Disorder Among Older Adults....
Naloxone [D009270]	Naloxone	Multiple Sclerosis [D009103]	Multiple Sclerosis	negative correlation	41143697	Secondary Trigeminal Neuralgia Due to Multiple Sclerosis in a Patient With Opioid Use Disorder Treat...
Naloxone [D009270]	Naloxone	Trigeminal Neuralgia [D014277]	Trigeminal Neuralgia	positive correlation	41143697	Secondary Trigeminal Neuralgia Due to Multiple Sclerosis in a Patient With Opioid Use Disorder Treat...

Naloxone [D009270]	Naloxone	Opioid-Related Disorders [D009293]	Opioid Use Disorder	negative correlation	41143697	Secondary Trigeminal Neuralgia Due to Multiple Sclerosis in a Patient With Opioid Use Disorder Treat...
Buprenorphine [D002047]	Buprenorphine	Multiple Sclerosis [D009103]	Multiple Sclerosis	negative correlation	41143697	Secondary Trigeminal Neuralgia Due to Multiple Sclerosis in a Patient With Opioid Use Disorder Treat...
Buprenorphine [D002047]	Buprenorphine	Trigeminal Neuralgia [D014277]	Trigeminal Neuralgia	negative correlation	41143697	Secondary Trigeminal Neuralgia Due to Multiple Sclerosis in a Patient With Opioid Use Disorder Treat...

The resulting network has 535 normalized entities and 530 edges supported by 2187 article-level evidence rows. Evidence is kept separate from aggregated edges so that a summary remains traceable to its relation label, PMID, mentions, and available sentence anchor. The article-level join changes display labels, not relation assertions: a valid MeSH UI receives its preferred thesaurus name, while the adjacent mention column records what PubTator3 actually linked in the article. Table 9 therefore also shows why this design matters. Some rows place clinically relevant concepts in an informative sentence, whereas others expose a relation label unsupported by the sentence or text misclassified as an entity. PubTator3 relations are model outputs (Wei et al., 2024), not validated causal claims; the evidence table makes review and filtering possible before a network is used substantively.

Figure 3 turns the evidence-preserving output into a deliberately focused display. It retains only same-sentence relations supported by at least two distinct PMIDs, excludes Species entities, and shows the one-hop neighborhoods of buprenorphine and OUD together with supported relations among their neighbors. Selection is based on article support rather than quotas for relation classes. Direction remains available in the underlying edge table, but arrowheads are suppressed because several displayed relation types are not naturally interpreted as directional claims.

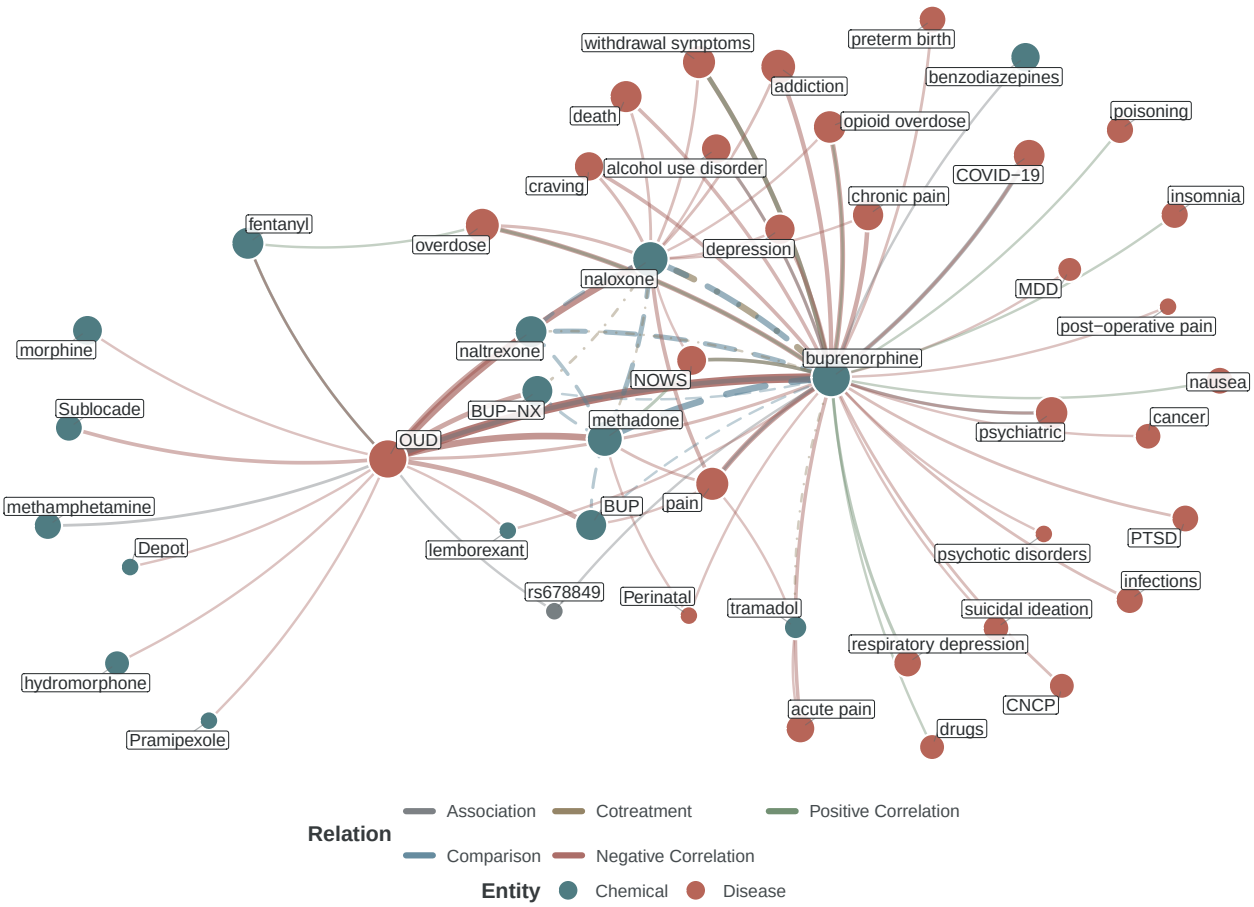


Figure 3. Supported PubTator3 relation neighborhood around buprenorphine and opioid use disorder (OUD). Nodes are entities and colors distinguish entity types; node size increases with the number of supporting PMIDs. Node labels are the most frequent PubTator3 surface form for each concept, whereas Tables 8 and 9 use MeSH preferred names. Edges are same-sentence PubTator3 relations supported by at least two distinct PMIDs; color and line type distinguish relation classes, and width increases with article support. Species entities are excluded. The display summarizes upstream model predictions and should not be interpreted as a validated biomedical or causal network.

The focal display makes the value of the returned graph structure visible: strongly supported links remain prominent, relation classes retain their observed frequency, and every edge can still be traced to the evidence rows used for Table 9. The figure is a downstream view of `pubtator_network()` output, not an additional extraction or graph-inference method.

4.5 Selective PMC full-text retrieval

The example ends with selected source documents. When abstract metadata and annotations are insufficient, seed PMIDs can be resolved to section-level full text. This optional step begins by resolving PMIDs to PMC open-data XML locations. Here, only the first 20 seed PMIDs are passed to the resolver and no more than three resolved articles are downloaded.

```
pmc_probe_pmids <- head(pmids, 20L)
pmc_map <- puremoe::pmid_to_ftp(pmc_probe_pmids, verbose = FALSE)
```

`pmid_to_ftp()` retains its original function name but, in the package version used here, converts eligible PMIDs to versioned XML URLs in PMC's public S3 cloud distribution rather than the retiring FTP/OA API (National Library of Medicine, 2026). The resolved URLs can then enter the same `get_records()` interface used elsewhere in the article. Table 10 shows the section-level text returned for the selected articles.

```
fulltext_urls <- head(pmc_map$url, 3L)
fulltext <- puremoe::get_records(
  fulltext_urls,
  endpoint = "pmc_fulltext",
  cores = 1L
)
```

Table 10. Representative section text from each retrieved PMC article.

pmid	section	text
42292651	Introduction	The opioid epidemic in the United States has been a significant public health issue [1]. In 2020, an estimated 2.5 million U.S. adults had opioid use disorder (OUD) [2], a chronic ...
42313747	Introduction	Chronic pain is commonly reported among people who use drugs (PWUD) [1], with approximately 48%–60% of people using prescription opioids non-medically, compared to 11%–19% among th...
42376697	Introduction	Opioid agonist medications buprenorphine and methadone are the most effective treatments for opioid use disorder (OUD), reducing overdose and HIV-risk behaviours and improving func...

Three files are sufficient to demonstrate that `pmc_fulltext` converts article XML into section-level text; this limit is pedagogical rather than an endpoint restriction. Table 10 normalizes whitespace and shortens displayed values. Section headings are returned separately in `section`; for display, a leading numeric or section label preserved in some source XML is removed when present. The complete returned `text` remains unchanged in `fulltext` for downstream use. No further text analysis is shown because this example is limited to section-level retrieval.

5. Implementation and performance

The implementation uses endpoint-specific adapters, source-specific local workflows, and conventional R data structures.

5.1 Endpoint adapters and execution

`get_records()` is a dispatcher over endpoint adapters rather than a common HTTP request with different URLs. In the package version used here, PubMed and iCite requests are divided into batches of at most 199 identifiers, PubTator3 requests into 99, and PMC full-text requests into five documents. Each adapter constructs the request expected by its service, parses the response, and returns a stable package-level

representation. PubMed XML becomes an article table with nested fields; iCite JSON becomes an article table with citation-link list-columns; PubTator3 BioC data become related entity and relation tables; and PMC's JATS-formatted XML from the public cloud distribution becomes section-level text (National Library of Medicine, 2026). BioC was created specifically to support portable biomedical text and annotation exchange (Comeau et al., 2013), but its document, passage, annotation, and relation hierarchy still requires an R representation suited to downstream analysis.

Independent batches can be evaluated on a user-selected number of parallel workers. The `sleep` argument remains exposed because parallel execution does not waive an upstream service's request policy. The delay is applied within each worker, so an approximate aggregate ceiling is `cores / sleep` requests per second; callers must choose both values together. The PubMed calls above use three workers and a 1.05-second delay to remain below NCBI's documented three-request-per-second limit even without a key. The iCite and PubTator3 calls used three workers with delays of 0.25 and 0.34 seconds, approximate aggregate ceilings of 12 and 9 requests per second; users should check each service's current guidance when choosing these values. NCBI API keys are passed through where applicable, and an iCite-specific timeout can be set by argument or package option. These controls are kept at the retrieval boundary; local functions do not need cores, credentials, sleeps, or retry settings. `endpoint_info()` records these endpoint differences alongside the input and return schemas so that the compact dispatcher does not conceal service-specific behavior.

5.2 Inspectable local workflows

The local workflows remove recurring source-specific wrangling without attempting to supply a general literature-analysis framework. Their functions take explicit retrieved objects as inputs, return tables rather than hidden state, and preserve enough source detail to trace a summary back to the records that produced it. The three workflow families apply that contract to different source structures.

`mesh_keyness()` compares corpora at the article level, not by counting repeated labels. It contrasts article-level descriptor prevalence with a PubMed-wide frequency baseline. The worked example ranks descriptors with a smoothed, standardized log-odds score. Positive values identify descriptors that are unusually prevalent in the selected corpus. These descriptive scores depend on the corpus and baseline version.

The two citation functions retain different links. `citation_snowball()` ranks records outside the seeds by their cited and citing links to the seed neighborhood. It then applies the requested direction, threshold, and `max_nodes` limit. `citation_network()` keeps only links whose endpoints occur inside the retrieved corpus. Both return the links underlying their summaries. Users can thus inspect admissions, change thresholds, or construct a graph without retrieving iCite data again. Edges follow citation flow: `from` is the citing PMID and `to` is the cited PMID. Accordingly, `cited_links` counts seed-to-reference edges and `citing_links` counts candidate-to-seed edges; `link_count` is the selected directional count or their sum. Network nodes are the PMIDs returned by iCite, including isolates, while edges are deduplicated links whose two endpoints both belong to that node set.

`pubtator_context()` restores textual context to upstream PubTator3 predictions. The sentence and offset alignment described in the worked example supports two outputs: co-occurrence summaries calculated from mapped entities and relation networks that aggregate PubTator3 predictions. The relation networks retain separate article- and sentence-level evidence rows. These rows allow users to review useful annotations and extraction errors before substantive interpretation.

5.3 Scale and interoperable representations

Remote work scales primarily with the number of endpoint-specific batches. Parallel workers can process independent batches, but useful concurrency remains bounded by service request policies, network availability, and the configured delay. Local workflows make no additional requests; their cost instead follows the number of article descriptors, citation edges, entity mentions, or relations in the retrieved object. PMC work is optional and potentially more request-intensive, so users should resolve only the subset selected for full-text analysis. These constraints make `puremoe` appropriate for moderate, investigator-scale corpora rather than reconstruction of PubMed or a global citation graph.

Most retrieval functions return `data.table` objects (Barrett et al., 2026) with PMIDs stored as character values. This avoids treating identifiers as quantities and permits ordinary joins, grouping, and export. List-columns preserve genuinely nested information, such as per-article MeSH annotations and iCite links, without duplicating article metadata. PubTator3 and network operations return named lists of tables when several related components are required. No hidden session object connects stages: the returned objects are the interface between retrieval, local workflows, and subsequent analysis.

The package likewise does not impose custom graph or visualization classes. `citation_network()` returns node and edge tables that can be passed directly to `igraph`, as shown above, or to another graph package. `puremoe` prepares the literature data while established R tools supply graphical style and statistical modeling. Intermediate results therefore remain usable even when the rest of an analysis does not use `puremoe`.

5.4 Reproducibility record

The submitted article renders from the dated object `data/puremoe-example.rds` by default. The accompanying `scripts/retrieve-data.R` script records the literal query and retrieval parameters, repeats every remote request displayed in the article, writes an endpoint return audit, and replaces the snapshot when a fresh live run is required. All transformations, tables, and figures are evaluated from the archived objects during the default build. This separates reproducibility of the reported results from availability and later changes in upstream services without concealing how the records were obtained.

Table 11. Environment and retrieval settings for the archived article snapshot.

Item	Value
Retrieval date	2026-09-11
puremoe version (retrieval)	1.1.0
puremoe version (render; CRAN)	1.1.1
R version	R version 4.3.1 (2023-06-16)
Platform	x86_64-conda-linux-gnu
Retrieval workers	3
NCBI API key used	Yes

The submission also includes `data/retrieval-sessionInfo.txt`, `data/render-sessionInfo.txt`, and `data/retrieval-audit.csv` as plain-text records. The first session record describes the original remote retrieval; the second describes the environment used for the submitted render and local workflows. A live build remains available by setting the `PUREMOE_LIVE_RETRIEVAL` environment variable to `true`, but it is expected to produce a later state of the upstream indexes rather than reproduce the dated counts exactly.

6. Limitations

`puremoe` depends on services that the package does not control. Availability, latency, schemas, rate limits, and indexing schedules may change. A valid live workflow can therefore be interrupted by an upstream outage, and the same query can yield different PMIDs on a later date. The article reports its retrieval date rather than presenting the current counts as fixed.

Saving retrieved objects can make downstream local workflows repeatable, but it does not make a changing external index historically immutable.

iCite links describe a PMID-based citation collection rather than a universal scholarly graph, and the PMC endpoint retrieves open-data XML rather than arbitrary publisher full text. These upstream boundaries also bound the local workflows that `puremoe` can support.

The local summaries reflect upstream metadata and automated annotations; they do not validate them. MeSH keyness depends on MEDLINE indexing and the packaged frequency baseline. Citation counts favor work with more time and opportunity to accumulate links, and expansion thresholds can exclude weakly connected but relevant articles. Same-sentence co-occurrence is not semantic association, and PubTator3 relation labels may contain entity or relation-extraction errors. Evidence rows support review but do not replace domain validation.

The package is intended for moderate, investigator-scale corpora. Parallel batching reduces wall-clock time but does not turn remote APIs into bulk data infrastructure; large or repeated analyses may be better served by local archives or bulk datasets. `puremoe` is not a PubMed mirror, a general interface to every NCBI database, a systematic-review management system, or a global scholarly graph. The worked query is deliberately precise and pedagogical. Its results demonstrate software behavior and corpus exploration, not an exhaustive search or synthesis of buprenorphine evidence.

7. Discussion

`puremoe` occupies a middle position between individual API clients and large bibliometric infrastructures. Its defining contribution is a reusable PMID corpus that can move through selected NIH/NLM services and then enter source-specific, inspectable local workflows. This supplies more workflow structure and biomedical-domain support than a generic request client while avoiding the storage and maintenance requirements of a literature mirror. In the worked example, each service response becomes an ordinary R object with an explicit relationship to the seeds.

The example also shows why combining these sources is analytically useful. MeSH descriptors summarize the seed corpus in a controlled vocabulary and link current language to older entry terms. Citation expansion then recovers highly connected trials, reviews, and guidelines outside the literal title query, including work framed as *opioid dependence*. The seed–candidate MeSH contrast describes how that expansion changes the corpus rather than merely enlarging it. PubTator3 adds a different view based on concepts in text, while retained sentence evidence exposes both potentially useful relations and obvious reasons for caution. A small PMC subset demonstrates sectioned full text without turning that optional endpoint into a corpus-wide download.

These gains occur at the transitions between services. PubMed annotations feed a comparison against a PubMed-wide baseline; iCite links support both a ranked candidate table and a graph; PubTator3 offsets become sentence context before evidence is aggregated. These local workflows go beyond returning upstream JSON, XML, or BioC, yet remain inspectable rather than forming an opaque end-to-end model.

The intended niche also defines when another package is preferable. `rentrez` is broader across NCBI; `pubmedR` offers a direct path from PubMed to `bibliometrix`; `europepmc` is the appropriate client when Europe PMC is the organizing index; and `openalexR` offers wider cross-disciplinary graph breadth. These clients can also be combined with `puremoe`, because an existing PMID vector can enter the workflow and returned tables use conventional R representations. The package is most useful when PMIDs are the organizing key and several NIH/NLM services are required in the same investigator-scale analysis.

Finally, live retrieval and reproducibility should not be conflated. Executable code, literal queries, dates, and parameters make the process transparent, but a future execution queries a future state of each service. Separating retrieval from local work provides a practical response: researchers can retain the retrieved tables associated with a reported analysis while freely rerunning local thresholds, networks, and visualizations. The archived article reproduces the reported analysis, whereas a fresh execution tests the current workflow. Empirical studies should also archive their analysis inputs where licensing and data volume permit.

8. Conclusion

`puremoe` turns a PubMed-derived or user-supplied PMID set into reusable R objects that can be enriched across selected NIH/NLM services and used in local workflows without hidden retrieval. The worked example showed the practical payoff: controlled vocabulary characterized the seeds, citation links exposed historically connected literature, PubTator3 predictions became traceable to sentence evidence, and a selected subset resolved to section-level PMC text.

The package is best suited to moderate, investigator-scale projects in which PMIDs are the organizing key and two or more of these services are needed in one R workflow. A single-service client is simpler when only one source is required; a broader scholarly graph or bulk archive is preferable when coverage extends beyond the NIH/NLM ecosystem or approaches database scale. Within its intended scope, the explicit retrieval boundary and conventional returned objects let users preserve the empirical corpus, rerun local workflows, and inspect the evidence behind each summary.

Software and Data Availability

puremoe 1.1.1 is available from CRAN (https://CRAN.R-project.org/package=puremoe); the source repository is https://github.com/jaytimm/puremoe. The article's reproduction materials comprise the dated retrieval snapshot (2026-09-11), the complete live-retrieval script, an endpoint return audit, and session information. All reported tables and figures render from the snapshot without contacting the upstream services. The article source and these materials are available at https://github.com/jaytimm/puremoe-paper.

Statement on the use of generative AI

Anthropic Claude Code was used to review and debug package and article code and to assist with manuscript revision and formatting. The author reviewed all generated output and takes full responsibility for the content.

Competing interests

The author has no competing interests to declare.

Authors' contributions

The author developed the package, designed and conducted the worked example, and wrote and revised the manuscript.

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